

RITIK KUMAR

5th-year, BS-MS (Research) Physics | Indian Institute of Science, Bengaluru
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PROFILE

Physics researcher at the Indian Institute of Science working in machine learning, statistical analysis, scientific computing, and large-scale data analysis. I have worked on developing machine-learning and probabilistic models for noisy and irregular time-series data, including Deep Gaussian Processes, ReFANN, Siamese networks, and Physics-Informed Neural Networks. My work involves model development, optimisation, uncertainty estimation, validation, and working with large datasets. I also have experience in astrophysical data analysis using XMM-Newton and NuSTAR. Strong background in probability, statistics, linear algebra, optimisation, and numerical methods, with most of my work done in Python. Strong mathematical foundations (JEE Advanced AIR OBC 3424, top 0.3%).

EDUCATION

Indian Institute of Science (IISc), Bengaluru

Aug 2022 – July 2027 (expected)

BS-MS (Research) in Physics

CGPA: 7.5/10

- Linear Algebra, Multi-variable Calculus, Probability & Statistics, Machine Learning, Matrix Theory, Stochastic models & Application, Mathematical Methods in Physics, Computational Physics. Radiative Processes, Galaxies & ISM, General Relativity & Cosmology, Astronomical Techniques, Fluid Mechanics & Plasma Physics

TECHNICAL SKILLS

- **ML / DL:** PyTorch, GPyTorch, Deep Gaussian Processes, ReFANN, Siamese Networks, PINNs, Bayesian Neural Networks, MC Dropout, uncertainty estimation
- **Quantitative / Statistical:** Probability and Statistics, Statistical Inference, Time-Series Analysis, Bayesian Inference, Quantile Regression, Cross-Validation, Bootstrap Resampling, Robust Statistics, MCMC
- **Mathematics:** Linear Algebra, Optimisation, Numerical Methods, Differential Equations, Dynamic Programming
- **Programming:** Python (NumPy, SciPy, Pandas, Astropy, scikit-learn, Matplotlib), Bash, some C, LaTeX.
- **Computing** HPC/SLURM, SSH, Parallel Processing, XMM-SAS, NuSTARDAS, XSPEC/ISIS

RESEARCH & ML EXPERIENCE

ML Researcher – GRB Light Curve Reconstruction

Apr 2024 – July 2026

National Astronomical Observatory of Japan (NAOJ) Advisor: Prof. Maria G. Dainotti

- Developed the complete Deep Gaussian Process, ReFANN, and Siamese-network models from scratch for reconstructing irregularly sampled and noisy GRB X-ray light curves.
- Contributed to building a large optical GRB dataset containing 535 GRBs and 64,813 observations, including data standardisation, automated GCN data collection, and analysis of optical colour evolution.
- Worked on difficult light-curve features such as plateaus, flares, and bumps, where simple interpolation methods can fail.
- Built the full pipeline from data preprocessing and log-space normalisation to model training, hyperparameter tuning, uncertainty estimation and validation.
- Implemented a two-layer Deep Gaussian Process using GPyTorch, with RBF kernels and variational inference, and applied it to more than 500 GRBs. The model reduced the uncertainty in $\log T_a$, $\log F_a$, and α by about 12%, 12%, and 16%, respectively.
- Developed ReFANN with three hidden layers, separate mean and log-variance outputs, Gaussian NLL loss, and a physics-inspired gradient regularisation term. It gave uncertainty reductions of 19.83%, 21.43%, and 26.51% for $\log T_a$, $\log F_a$, and α , respectively, across 522 GRBs.
- Developed a Siamese network with shared weights and a learnable reference anchor to capture both absolute and relative temporal information. Used MC Dropout to estimate predictive uncertainty and obtained uncertainty reductions of 22.88%, 22.83%, and 26.93% for $\log T_a$, $\log F_a$, and α .
- Performed five-fold cross-validation, bootstrap analysis, ablation studies, MC-Dropout convergence tests and outlier analysis to check the stability of the models.
- Worked on comparing different modelling approaches based not only on prediction accuracy but also on uncertainty and robustness. Our study compares PINN, ReFANN, Siamese NN and Polynomial Quantile Regression on 545 GRBs.

- **Published:** 1. Kaushal, Dainotti, Ritik et al. (2026), and 2. Ayush, Ritik, Dainotti et al. (2026-arxiv pre-print and it is in (peer-review in journal). 3. M G Dainotti,,,,Ritik et al.(2024), Code publicly available: [Github](#).

ML Researcher – Physics-Informed Neural Networks (PINNs)

Jan 2025 – Apr 2025

IISc Bengaluru Advisor: Prof. Rishita Das (Aerospace Engineering)

- Implemented PINNs in PyTorch to solve 1D, 2D, and 3D Burgers' equations and 1D Euler equations, including the Sod shock-tube problem with sharp discontinuities.
- Compared the results with Runge-Kutta, Euler, central-difference methods, used CFL criterion for stable time-stepping.

Lead Researcher – AGN Spectral Modelling (ECAP Collaboration)

Jan 2026 – Present

Erlangen Centre for Astroparticle Physics (ECAP), FAU Germany Full Group Member

- Working with XMM-Newton and NuSTAR multi-epoch data for X-ray spectral analysis of active galactic nuclei.
- Test relativistic reflection models and use χ^2 fitting, MCMC and model comparison to study the X-ray spectra.
- Developed Python-based data-reduction and analysis pipelines and run the analysis on the Remeis HPC cluster using parallel processing.

Tracing of Outflows in Active Galactic Nuclei – Bachelor's Thesis

Sep 2025 – Apr 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Built a Python pipeline to cross-match about 1 million AGN catalogue entries with XMM-Newton observations and automate filtering and spectral extraction.
- Used 23-worker parallel processing and developed custom XSTAR grids to study complex X-ray absorption features and multi-phase ultra-fast outflows. **Manuscript in preparation.**

Spectral Analysis of Seyfert Galaxies

Dec 2024 – Jan 2026

Indian Institute of Astrophysics (IIA) Advisor: Prof. C. S. Stalin

- Developed a Python pipeline to cross-match 30,000 sources, perform data reduction, background and S/N selection.
- Worked with NuSTAR and XMM multi-epoch observations and performed broadband spectral modelling in XSPEC.
- Used X-ray spectral modelling to study and measure coronal temperatures in Seyfert galaxies.
- **Manuscript in preparation.**

PUBLICATIONS

Publications

- Ayush, **Ritik**, Dainotti et al. (2026). Physics-Informed Neural Networks and Data-Driven Models for GRB X-ray Light-Curve Gap Reconstruction. [2026](#). *arXiv pre-print, paper is in peer-review*.
- Kaushal, A., Dainotti, M.G., **Ritik**, et al. (2026). Multi-Model Framework for Reconstructing Gamma-Ray Burst Light Curves. *Journal of High Energy Astrophysics (JHEAp)*. [2026](#)
- Dainotti,,,,**Ritik** et al. (2024). An optical gamma-ray burst catalogue with measured redshift – I. Data release of 535 gamma-ray bursts and colour evolution. *MNRAS Journal*. [2024](#)

In Preparation

Ritik, Stalin, C.S., Reeves, J.N. Possible detection of unambiguous outflows in a Luminous Quasar. In preparation, 2026.

Ritik, Stalin, C.S., Hota, J. Variation in Coronal Temperature in Seyfert-type AGN. In preparation, 2026.

CONFERENCES & WORKSHOPS

- XRISM International Conference 2025, Japan – [Certificate](#).
- Astronomical Society of India Conference 2026, Guwahati – ([Certificate](#), [Poster](#)).

ACHIEVEMENTS

- **Internship, NAOJ Tokyo (Summer 2025)** and **collaboration** with Prof. Maria G. Dainotti since April 2024.
- **Full membership, ECAP X-ray Research Group, Germany** since Jan. 2026
- **Research collaboration with the Indian Institute of Astrophysics (IIA), Bengaluru** since Dec. 2024.
- **Management Coordinator**, Pravega 2024–25 (India's largest science & cultural festival), IISc